

Die chipping EP3C25T60UC256N

8D Report

20 Apr 2017

Revision: 2

1D – Team Member

Appendix _ 8D Report

	Commento			
	Name	Department	Email ID	
Team Lead	Lee	PE		
Team Member	Gwon	PE		
	Moon	ATK TPM		
	Yun	QE		
	Kim	Production		

	Altera			
	Name	Department	Email ID	Phone #
Main Contact	Joo			



2D – Problem Description

Appendix _ 8D Report

Problem Statement	
<i>Die chipping phenomenon was detected after wafer sawing process</i>	

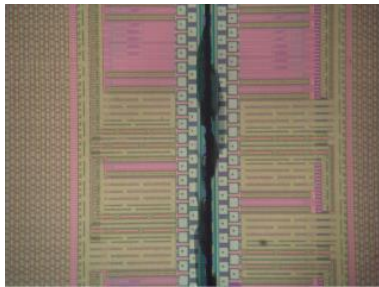
Problem Detail			
Altera CAR #	N/A	Defect mode	<i>Die chipping</i>
Altera MDP #	N/A	Impacted Lot # (qty)	69381 69382
Altera ERMA #	N/A	Total impacted qty	1968
Supplier 8D #		Date code	
Assy / Test Site	ATK4	Detection point	ATK4
Device #	EP3C25T60	Detection date	12/04/2017
Package type	PBGA	Detection process	Wafer saw
Reject qty	809	Occurrence date	12/04/2017
Total qty inspected	1968	Occurrence process	Wafer saw
Reject rate	41.1%		



2D – Problem Description

Description of defect phenomenon

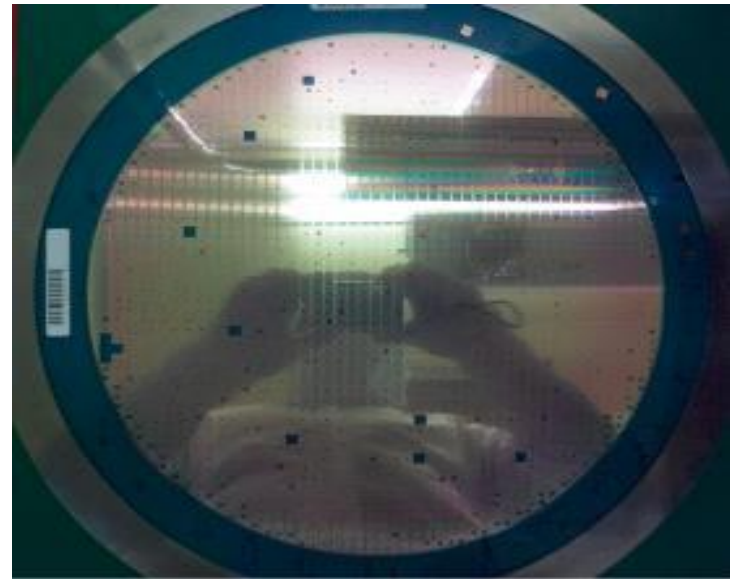
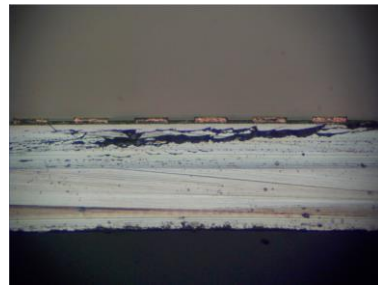
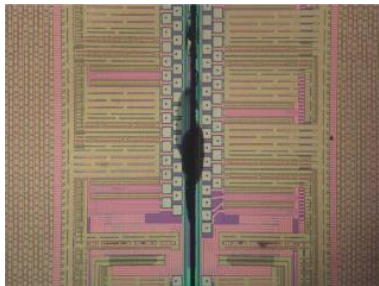
- Die chipping phenomenon was detected after wafer sawing process
- Done 100% inspection by AOI and high power scope and 809 dies were affected (Affected yield : 41.1%)



Top



Lateral



Affected area

3D – Containment Action

Appendix _ 8D Report

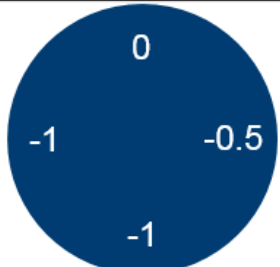
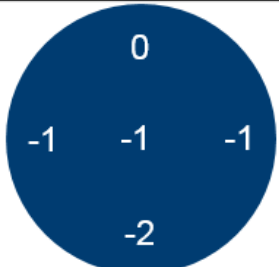
#	Action (Amkor)	Due Date	Owner
1	100% inspection for affected 4 PCS wafers thru high power scope & AOI machine Result : Done (reflected reject position on the wafer)	Done 17/04/2017	Kim – MFG & Gwon –PE
2	Check condition of continuously sawing wafer (Same device but change the blade) Result: No abnormality is confirmed	Done 14/04/2017	Kim – MFG & Gwon –PE
3	Do not use affected machine for ALTERA device temporary Result : No abnormality is confirmed	Done 15/04/2017	Kim – MFG



3D – Containment Action

Check the sawing machine

- Check the machine vibration, flange condition and chuck table flatness
 - Any abnormality was not detected and all item meet the spec criteria

Accuracy	
Base	Cutting table
Spec criteria : 3um	
	

Mode	Vibration SPEC	
	40000rpm	50000rpm
Item	1.5um	2um
SP1 Disp.EQP	N/A	0.07um
SP2 Disp.EQP	0.08um	N/A

Item	Criteria	Result
SP1 Flange Condition	Z1) $\leq 0.003\text{mm}/360^\circ$	1um
SP2 Flange Condition	Z2) $\leq 0.003\text{mm}/360^\circ$	0.5um
SP1 Spindle Tip Axial Runout	Z1) $\leq 0.0005\text{mm}/360^\circ$	0.5um
SP2 Spindle Tip Axial Runout	Z1) $\leq 0.0005\text{mm}/360^\circ$	0.5um

3D – Containment Action

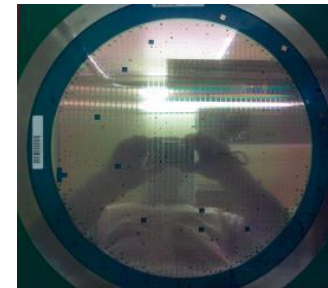
Result of inspection for affected wafer

- During AOI inspection, 505 dies were classified as reject dies
- Sorted dies which is detected during AOI process was already classified as reject dies during high power cope inspection

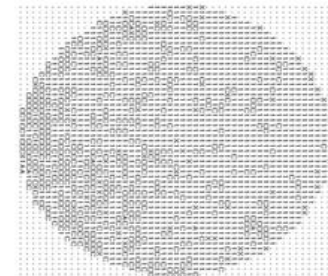
100% inspection for next sawn wafer

- Any chipping or abnormality was not detected at the next sawing wafer

Customer	PDL	Device	Lot #	Remark & Inspection result
ALTERA	1A-14-256	EP3C25T60UC256N	NCAAA69382	Affected wafers
Z1, Z2 blade is changed				
ALTERA	1J-19-484	EP3C55T60UI484	NCBAA68449	No chipping or abnormality
ALTERA	1J-19-484	EP3C55T60UI484	NCBAA68449	No chipping or abnormality
ALTERA	1A-14-256	EP3C25T60UI256VN	NCAAA68717	No chipping or abnormality



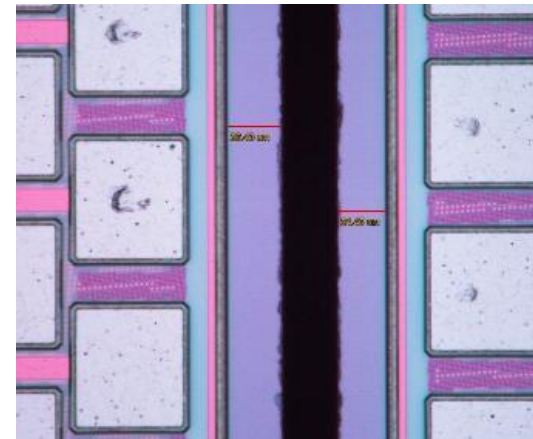
High power inspection



AOI inspection

4D – Root Cause Analysis - Device informatio

- Device Code : Q9 / 69381
1A / 69382
- Device : EP3C25T60
- Saw street width : Around 80um
- Use blade kerf width : Z1(Max 30um), Z2(Max 25um)



4D – Root Cause Analysis

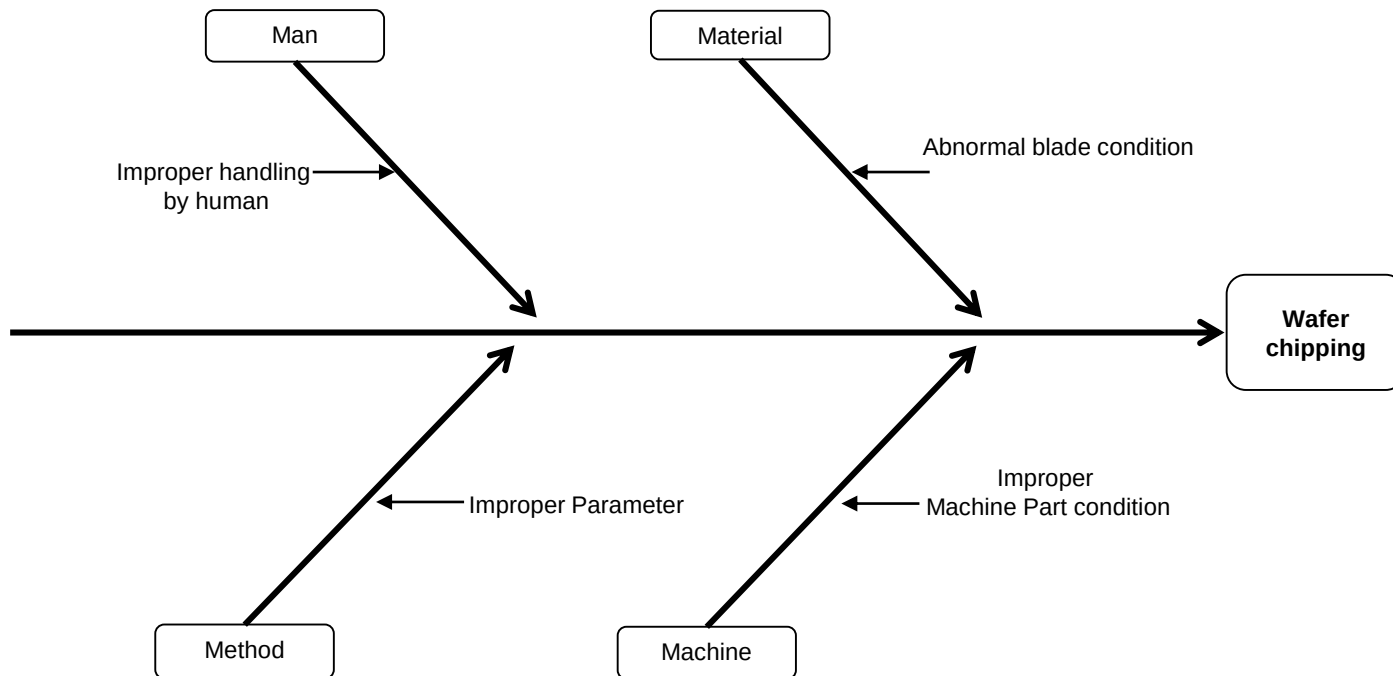
Commonality Study for affected lot

- Saved recipe was used for affected wafer and any abnormal machine error was not happened
 - Remained blade exposure is enough to saw the affected wafer
(Spec criteria : Blade exposure should be over than 2.5mil, Remained blade exposure : 10mil)
- Any chipping or abnormality was not detected at the next lots
 - Apply same machine for next lot but after change the blade, any abnormality was not detected

Customer	PDL	Device	Lot #	Date	Saw Speed	Spindle RPM	Blade height	Z1 exposure	Z2 exposure	Blade
Z1, Z2 change										
Other	*	*	*	2017-04-12 1:03	0.7	50000 / 35000	5.96 / 3.4	24.61	19.53	SD3500-CC / 27HCBB
Z2 change										
ALTERA	1A-14-256	EP3C25T60UC256N	NCAAA69382	2017-04-12 16:38	0.5	50000 / 35000	8.3 / 2.7	23.54	19.56	SD3500-CC / 27HCCB
Z1, Z2 change										
ALTERA	1J-19-484	EP3C55T60UI484	NCBAA68449	2017-04-13 5:08	0.5	50000 / 35000	9.2 / 2.7	24.77	24.49	SD3500-CC / 27HCCB
ALTERA	1J-19-484	EP3C55T60UI484	NCBAA68449	2017-04-13 6:05	0.5	50000 / 35000	9.2 / 2.7	24.64	24.44	SD3500-CC / 27HCCB
ALTERA	1A-14-256	EP3C25T60UI256VN	NCAAA68717	2017-04-13 7:34	0.5	50000 / 35000	9.2 / 2.7	24.51	24.35	SD3500-CC / 27HCCB
ALTERA	1J-19-484	EP2C35Y90UC484-8N	VBD9Y68411	2017-04-13 10:23	0.5	50000 / 35000	9.2 / 2.7	24.51	24.35	SD3500-CC / 27HCCB
ALTERA	1A-14-256	EP3C10T60UI256VN	NCAAA69076	2017-04-13 14:58	0.5	50000 / 35000	9.2 / 2.7	24.19	24.27	SD3500-CC / 27HCCB

4D – Root Cause Analysis

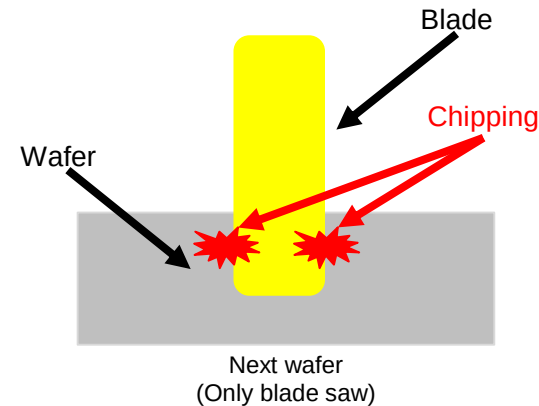
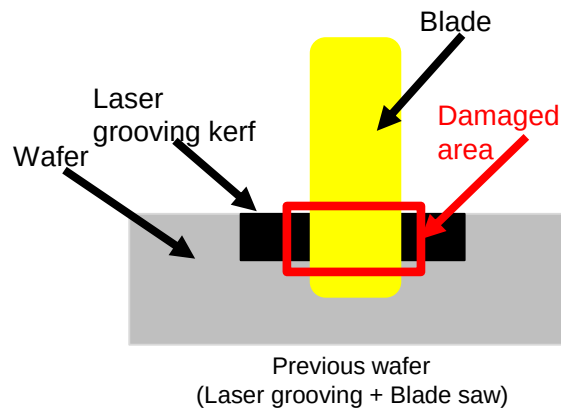
Ishikawa's fish bone analysis



4D – Root Cause Analysis

Blade clogging or damage by debris

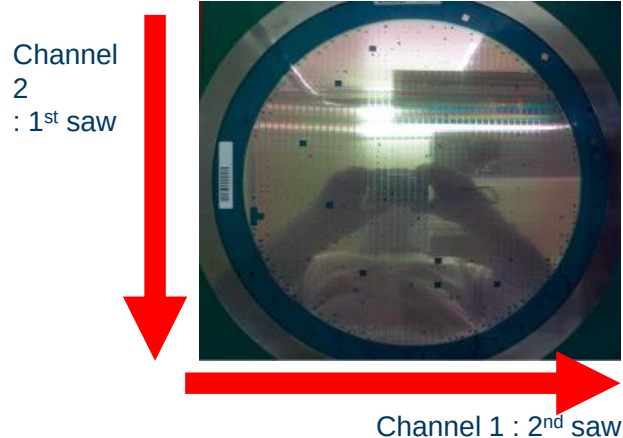
- Laser debris is made because metal on the saw street is massed during laser grooving process and it is remained at the saw street
- Laser debris can occur the clogging or damage on the surface of blade
- And damaged blade can give the stress to wafer and make chipping phenomenon

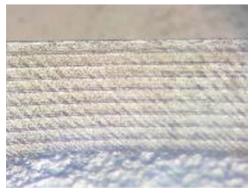
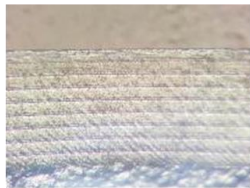
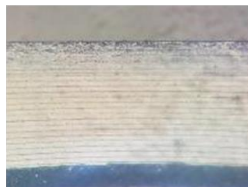


4D – Root Cause Analysis

Blade clogging or damage by debris

- Channel 2 is sawn first after that channel 1 is sawn second
 - Wafer sawing is started channel 2 -> Done channel 2 -> Wafer sawing is started channel 1 -> Done channel 1
- Most chipping phenomenon was detected at the channel 2 and kerf condition of channel 1 is not bad
- But any blade damage or abnormality was not detected at the surface of blade due to blade condition is returned to normal because shelf sharpening effect when proceed the channel 2 area



	Blade surface	
Z1 Blade		
Z2 Blade		

4D – Root Cause Analysis

Summary of findings

- Before the affected wafer is sawn, blade was used for laser grooved wafer
- And after change the blade, any chipping or abnormality was not detected even apply same parameter and machine

Conclusion

- Because affected blade was used for laser grooved wafer, metal clogging was happened at the surface of blade
- So wafer is taken the damage and chipping phenomenon was occurred

4M Risk Assessment

4M	Possible cause	Assessment	Risk
Man	Improper handling by human	No abnormality	Low
Material	Abnormal blade condition	Laser grooved wafer is sawn using affected blade before the ALTERA device	High
Method	Improper parameter	No machine abnormal record is confirmed	Low
Machine	Improper machine part condition	No machine abnormal record is confirmed	Low

4D – Root Cause Analysis

Why – Why Analysis

Potential Root Cause
<i>Wafer chipping</i>
<i>it a symptom or root cause? Proceed to next Why if it is a symptom</i>
<i>Abnormal blade condition</i>
2 nd – Why did this happen?
<i>Metal clogging at the blade surface</i>
3 rd – Why did this happen?
<i>Laser debris is stitched at the blade surface and it isn't removed</i>
4 th – Why did this happen?
<i>Blade dressing was not applied before proceed the wafer saw process for affected wafer</i>
5 th – Why did this happen?
<i>NA</i>



5D – Corrective Action

Appendix _ 8D Report

#	Action	Due Date	Owner
1	Apply the blade dressing if blade is continuously used after laser grooved wafer sawing	Done	Kim - MGF
2	Apply the auto dressing for ALTERA specific device	Done	Gwon - PE



6D – Verification

Appendix _ 8D Report

#	Action	Due Date	Owner
1	Enhanced monitoring for next input lots _ No concern	Done	Gwon
2	Device RPM Sequence Speed Height Blade 1st evaluation 2nd evaluation EP3C10T60 50000 / 35000 CH2 -> CH1 0.5 IPS 8.3/2.7 (50%) SD3500CC /27HCCB No chipping TBD EP3C16T60 50000 / 35000 CH2 -> CH1 0.5 IPS 9.2/2.7 (40%) SD3500CC /27HCCB Chipping (Auto dressing -> Chipping) Chipping (Auto dressing -> Chipping -> No auto dressing -> No chipping -> Auto dressing -> Chipping) EP3C25T60 50000 / 35000 CH2 -> CH1 0.5 IPS 9.2/2.7 (40%) SD3500CC /27HCCB No chipping No chipping (PLS update) EP4CGX15T60 50000 / 35000 CH2 -> CH1 0.5 IPS 8.7/2.7 (45%) SD3500CC /27HCCB No chipping TBD Spec criteria Max 60000 N/A Max 4.0 IPS 10~80% / Full cut N/A N/A N/A	Done	Gwon



Evaluation

D7 - Recurrence Prevention

Item	Yes/No	Evidence if Yes																																																																																																																																																																																																																												
Fan-out	Yes	Update the spec of saw blade dressing (Spec # 001-2882)																																																																																																																																																																																																																												
FMEA	Yes	<table><tr><td colspan="10">Failure Mode and Effects Analysis (Process FMEA)</td><td colspan="2">FMEA NO: ATK-CASDA-21-0188</td><td colspan="2">PREPARED BY: PE</td><td colspan="2">DATE: 08/08/2021</td><td colspan="2">BY: JFJ</td><td colspan="2">FOR: JFJ</td></tr><tr><td colspan="10">FMEA DATE (original): 2020-10-30</td><td colspan="2">FMEA DATE (revised): 2021-04-14</td><td colspan="2">ACTION RESULTS</td><td colspan="2">ACTION TAKEN</td><td colspan="2">REV</td><td colspan="2">DOC</td><td colspan="2">DET</td><td colspan="2">WPK</td></tr><tr><td colspan="2">FUNCTION</td><td colspan="2">POTENTIAL FAILURE MODE</td><td colspan="2">POTENTIAL EFFECT OF FAILURE</td><td colspan="2">SEV</td><td colspan="2">CAUSE OF FAILURE</td><td colspan="2">SEV</td><td colspan="2">CURRENT PROCESS CONTROL</td><td colspan="2">CURRENT PROCESS CONTROL</td><td colspan="2">SEV</td><td colspan="2">WPK</td><td colspan="2">RECOMMENDED ACTIONS</td><td colspan="2">RE SP</td><td colspan="2">THNG</td><td colspan="2">ACTION RESULTS</td><td colspan="2">ACTION TAKEN</td><td colspan="2">REV</td><td colspan="2">DOC</td><td colspan="2">DET</td><td colspan="2">WPK</td></tr><tr><td colspan="2">Requirements</td><td colspan="2">Saw chip (kerf width)</td><td colspan="2">Electrical failure (Function, OS)</td><td colspan="2">7</td><td colspan="2">Improper saw parameter</td><td colspan="2">2</td><td colspan="2">1. PLS</td><td colspan="2">1. Visual inspection 2. Auto kerf check *</td><td colspan="2">2</td><td colspan="2">25</td><td colspan="2">None</td><td colspan="2"></td><td colspan="2"></td><td colspan="2"></td><td colspan="2"></td><td colspan="2"></td><td colspan="2"></td><td colspan="2"></td></tr><tr><td colspan="2">/ No visual fail No ESD fail No OS fail</td><td colspan="2"></td><td colspan="2"></td><td colspan="2"></td><td colspan="2">Blade clogging</td><td colspan="2">3</td><td colspan="2">1. Auto dressing 2. Auto Override 3. Blade dressing after chipping sensitive device</td><td colspan="2">1. Visual inspection 2. Auto kerf check *</td><td colspan="2">3</td><td colspan="2">55</td><td colspan="2">Apply the blade dressing after blade is used for laser grooved wafer</td><td colspan="2">MFG</td><td colspan="2">17.07.01</td><td colspan="2">Update the spec for saw blade dressing</td><td colspan="2"></td><td colspan="2"></td><td colspan="2"></td></tr><tr><td colspan="2"></td><td colspan="2"></td><td colspan="2"></td><td colspan="2"></td><td colspan="2">Improper parameter for 3mil wafer sawing</td><td colspan="2">2</td><td colspan="2">1. PLS 2. Parameter optimization</td><td colspan="2">1. Visual inspection 2. Auto kerf check *</td><td colspan="2">2</td><td colspan="2">25</td><td colspan="2">None</td><td colspan="2"></td><td colspan="2"></td><td colspan="2"></td><td colspan="2"></td><td colspan="2"></td><td colspan="2"></td></tr><tr><td colspan="2"></td><td colspan="2"></td><td colspan="2"></td><td colspan="2"></td><td colspan="2">Blade clogging due to previous wafer condition</td><td colspan="2">2</td><td colspan="2">1. Auto dressing 2. Auto Override 3. Blade dressing after chipping sensitive device</td><td colspan="2">1. Visual inspection 2. Auto kerf check *</td><td colspan="2">2</td><td colspan="2">25</td><td colspan="2">None</td><td colspan="2"></td><td colspan="2"></td><td colspan="2"></td><td colspan="2"></td><td colspan="2"></td><td colspan="2"></td></tr></table>	Failure Mode and Effects Analysis (Process FMEA)										FMEA NO: ATK-CASDA-21-0188		PREPARED BY: PE		DATE: 08/08/2021		BY: JFJ		FOR: JFJ		FMEA DATE (original): 2020-10-30										FMEA DATE (revised): 2021-04-14		ACTION RESULTS		ACTION TAKEN		REV		DOC		DET		WPK		FUNCTION		POTENTIAL FAILURE MODE		POTENTIAL EFFECT OF FAILURE		SEV		CAUSE OF FAILURE		SEV		CURRENT PROCESS CONTROL		CURRENT PROCESS CONTROL		SEV		WPK		RECOMMENDED ACTIONS		RE SP		THNG		ACTION RESULTS		ACTION TAKEN		REV		DOC		DET		WPK		Requirements		Saw chip (kerf width)		Electrical failure (Function, OS)		7		Improper saw parameter		2		1. PLS		1. Visual inspection 2. Auto kerf check *		2		25		None																/ No visual fail No ESD fail No OS fail								Blade clogging		3		1. Auto dressing 2. Auto Override 3. Blade dressing after chipping sensitive device		1. Visual inspection 2. Auto kerf check *		3		55		Apply the blade dressing after blade is used for laser grooved wafer		MFG		17.07.01		Update the spec for saw blade dressing																Improper parameter for 3mil wafer sawing		2		1. PLS 2. Parameter optimization		1. Visual inspection 2. Auto kerf check *		2		25		None																						Blade clogging due to previous wafer condition		2		1. Auto dressing 2. Auto Override 3. Blade dressing after chipping sensitive device		1. Visual inspection 2. Auto kerf check *		2		25		None													
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8D – Resolution

Appendix _ 8D Report

All team member for the effort					
Lee – PE MGR, Yun - QE, Kim - MFG MGR, Gwon - PE					
8D report approval					
Name	Yang	Position	QE Part Manager	Date	May.02.2017

